Ch-20 Engineering Economy & Project Management Homework Answers

01

The Safety Department intends to purchase 100 new lanyards & harnesses in 3 years at a cost of $11,000.00. How much should be invested (annual interest rate equals 4.0%) to cover the $11,000.00 cost in 3 years? You will need the following formula: $P = F\,(1+i)^{-n}$

A. $8,898.72

B. $9,335.87

C. $9,778.96

D. $10,040.44

$$P = \$11{,}000\ (1 + 0.04)^{-3} = \$9{,}778.96$$

02

What annual payment is required for a 10-year loan of $500,000 with a 7% interest rate? You will need the following formula:

$$A = P\left(\frac{i(1+i)^n}{(1+i)^n - 1}\right)$$

A. $25,343.82

B. $55,678.18

C. **$71,354.71**

D. $80,000.00

$i = 0.07$
$n = 10$
$P = \$500{,}000$

$$A=\$500{,}000\left(\frac{0.07(1+0.07)^{10}}{(1+0.07)^{10}-1}\right)$$

$$A = \$500{,}000\left(\frac{0.138}{0.967}\right) = \$71{,}354.71$$

03

If you invest $10,000 in an exoskeleton for workers who must attach bolts from an assembly project and it saves $15,000 annually in medical injuries, what is the ROI as a percentage?

A. **150%**

B. 50%

C. 15%

D. 5%

04

An investment of $20,000 is expected to earn an annual interest rate of 2.5%. What is the approximate future value of this investment after 3.5 years? $F = P(1+ i)^n$

A. $23,610.37

B. $20,902.69

C. **$21,805.37**

D. $25,000.69

$F = \$20{,}000\ (1 + 0.025)^{3.5}$

$= \$21{,}805.37$

05

A new ventilation system will cost $1,200,000 in five years. How much money should a company invest at 4.5% annual interest to reach this number? $F = P(1 + i)^n$

A. $120,000

B. $1,052,631

C. $1,114,857

D. $962,964

$$\$1,200,000 = P(1 + 0.045)^5$$

$$P = \$1,200,000 / 1.246$$

$$P = 962,964$$

06

What is a project’s scope?

A. **The work required to initiate and sustain a project towards completion.**

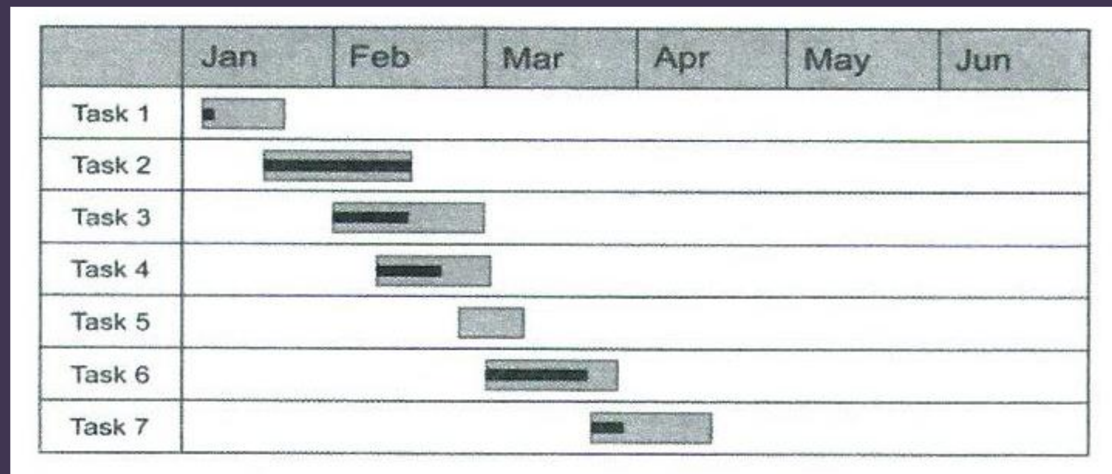
B. Only the work required to complete the project successfully

C. The work required by the project manager to fulfill the work breakdown structure

D. Only the work defined in the project’s charter

07

What type of project management tool is shown?



A. Cleo chart

B. **Gantt chart**

C. PERT chart

D. CAMEO chart

08

A ratio of a project or countermeasure input cost to a project or countermeasure impact measure, that is, dollars spent per accident prevented, is called:

A. Input / Output Analysis

B. **Cost-Effectiveness Ratio**

C. Cost-Benefit Ratio

D. Cost-Countermeasure Analysis

09

The ratio of the risk exposure to the project cost is called:

A. Loss ratio

B. Cost ratio

C. **Risk exposure ratio**

D. Risk ratio

10

What is risk mitigation in project management?

A. The process of transferring all project risks to another party to handle.

B. The act of ignoring minor risks until they become significant.

C. The process of taking proactive steps to reduce the likelihood or impact of potential project risks.

D. The practice of insuring against all possible risks to ensure no financial loss